

EKF vs. Shampoo-Powered Batch Smoother for Attitude Estimation

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Abstract: Attitude estimation from inertial measurement unit (IMU) data is a fundamental problem in robotics and autonomous flight. The extended Kalman filter (EKF) is the classical solution, but its recursive structure forces a single linearization per step, discarding raw measurements after each update and permanently locking in linearization errors that compound over dynamic maneuvers. We pose attitude estimation as a nonlinear least squares problem over a batch of gyroscope measurements on $SO(3)$ and solve it using Shampoo, a second-order preconditioned optimizer that exploits the matrix structure of the trajectory optimization variable. We compare both methods on the EuRoC MAV dataset across sequences of varying difficulty and evaluate using geodesic attitude error against motion capture ground truth. Our results confirm that a globally optimized trajectory initialized from ground truth achieves over $100\times$ lower geodesic attitude error than the recursive EKF on both sequences, at the cost of an offline requirement and moderately higher runtime, with the gap widening on aggressive motion as EKF linearization errors compound.

1 Introduction

Accurate attitude estimation, determining the orientation of a rigid body in three-dimensional space, is a foundational requirement in robotics and autonomous systems. For aerial vehicles in particular, reliable real-time attitude estimates are essential for stable flight control, trajectory planning, and state estimation applications. Inertial measurement units (IMUs) are the primary sensor used for this purpose, providing high-rate gyroscope and accelerometer measurements that can be integrated to track orientation over time.

The extended Kalman filter (EKF) has been the standard approach for IMU-based attitude estimation. Its recursive predict-update structure is computationally efficient and well-suited to real-time operation. However, the EKF carries a fundamental limitation: at each timestep it linearizes the nonlinear dynamics around the current estimate, discards the raw measurement, and propagates only a compressed summary of the past in the form of a mean and covariance. This means that linearization errors made early in the sequence cannot be corrected later, and during aggressive maneuvers where the linearization assumptions break down the most, these errors compound. The EKF update is equivalent to performing a single step of Gauss-Newton on the local linearized MAP objective, iterating to convergence in exchange for computational efficiency.

Batch optimization offers an alternative perspective. Rather than processing measurements recursively, a batch smoother collects the full measurement sequence and solves for the orientation trajectory that best explains all of them simultaneously. This formulation naturally allows the optimizer to re-linearize around progressively better estimates at each iteration, correcting mistakes that a recursive filter would lock in permanently. Both approaches can be understood as optimizing the same underlying gyroscope consistency objective on $SO(3)$, the EKF solves it approximately and recursively with Vicor attitude corrections at each step, while the batch smoother solves it globally and iteratively, initialized from the ground truth trajectory to provide the absolute pose reference that

the gyro-only cost function cannot supply on its own. The implications of this design choice are discussed in Section 5.

Because orientation lives on the special orthogonal group $SO(3)$ rather than Euclidean space, standard optimization methods cannot be applied directly. We handle this by optimizing perturbations in the tangent space of $SO(3)$ and retracting back to the manifold after each step via the exponential map, following standard practice in geometric state estimation [1].

In this work, we implement and compare two attitude estimators on real IMU data from the EuRoC MAV dataset [2]: an error-state EKF following the formulation of Solà [1], and a full-batch smoother minimizing a nonlinear least squares cost on $SO(3)$ using Shampoo [3], a second-order preconditioned optimizer, as the solver. We evaluate both methods across sequences of varying difficulty and measure performance using geodesic attitude error against motion capture ground truth. On easy motion both methods are competitive, while on aggressive sequences the batch smoother substantially outperforms the EKF as linearization assumptions break down and errors compound.

2 Related Work

Error-state EKF for attitude estimation. The error-state Kalman filter (ESKF) is the standard recursive estimator for orientation tracking on $SO(3)$. Rather than maintaining the full rotation matrix as the filter state, the ESKF parameterizes uncertainty as a small perturbation in the tangent space $\mathbb{R}^3$, where linearization is well-conditioned. Solà [1] provides a comprehensive interpretation of this formulation, deriving the predict and update equations for IMU-driven attitude estimation using both quaternions and rotation matrices. We use this formulation directly as our recursive baseline.

Batch smoothing for visual-inertial odometry. Leutenegger et al. [4] demonstrate that framing visual-inertial state estimation as a nonlinear least squares problem solved over a keyframe window consistently outperforms filtering-based approaches, particularly during aggressive motion. Their work establishes the conceptual foundation for our batch smoother, rather than processing measurements recursively, collecting them and optimizing jointly allows re-linearization around progressively better estimates. We apply this same batch optimization philosophy to the attitude-only problem, replacing their visual-inertial cost function with a purely gyroscope-driven residual on $SO(3)$.

Shampoo optimizer. Gupta et al. [3] introduce Shampoo, a second-order preconditioned optimizer that approximates the full-matrix AdaGrad preconditioner via a Kronecker product of two smaller matrices. For a matrix-shaped parameter $W \in \mathbb{R}^{m \times n}$, Shampoo maintains left and right preconditioner matrices $L \in \mathbb{R}^{m \times m}$ and $R \in \mathbb{R}^{n \times n}$ and performs the update $W \leftarrow W - \eta L^{-1/4} G R^{-1/4}$, where G is the gradient. This is substantially cheaper than full Newton while retaining second-order curvature information. Our optimization variable (perturbations to the rotation trajectory over T timesteps) is naturally matrix-shaped with time along one axis and tangent space dimensions along the other, making Shampoo a natural fit.

EuRoC MAV dataset. Burri et al. [2] present the EuRoC MAV dataset, a collection of visual-inertial sequences recorded onboard a micro aerial vehicle across environments of varying difficulty. The dataset provides synchronized IMU measurements at 200 Hz and millimeter-accurate pose ground truth from a Vicon motion capture system, making it a standard benchmark for state estimation algorithms. We use the IMU gyroscope data and Vicon ground truth exclusively, ignoring the camera data, as our focus is attitude-only estimation.

3 Methods

3.1 Problem Formulation

Let $R_t \in \text{SO}(3)$ denote the orientation of the vehicle body frame relative to a fixed world frame at time t . Given gyroscope measurements $\omega_t \in \mathbb{R}^3$ at rate Δt , the discrete-time rotation kinematics are:

$$R_{t+1} = R_t \cdot \text{Exp}(\omega_t \Delta t) \quad (1)$$

where $\text{Exp} : \mathbb{R}^3 \rightarrow \text{SO}(3)$ is the exponential map. Both methods optimize the same gyroscope consistency objective over the rotation trajectory $R_{0:T}$:

$$\min_{R_{0:T}} \sum_{t=0}^{T-1} \|\text{Log}(R_{t+1}^\top R_t \cdot \text{Exp}(\omega_t \Delta t))\|^2 \quad (2)$$

where $\text{Log} : \text{SO}(3) \rightarrow \mathbb{R}^3$ is the logarithmic map. Each residual measures the geodesic discrepancy between the propagated rotation and the next estimated rotation. The EKF solves this recursively with a single Gauss-Newton step per measurement and Vicon attitude corrections at each step; the batch smoother solves it globally using Shampoo, initialized from the ground truth trajectory.

3.2 SO(3) Parameterization

Since $\text{SO}(3)$ is a smooth manifold, we cannot optimize rotation matrices directly without violating the orthogonality constraint $R^\top R = I$, $\det(R) = 1$. We instead optimize a perturbation $\delta\phi_t \in \mathbb{R}^3$ in the tangent space at each timestep and retract back to the manifold after each update:

$$R_t \leftarrow R_t \cdot \text{Exp}(\delta\phi_t) \quad (3)$$

After retraction, $\delta\phi_t$ is reset to zero and the process repeats. This is the standard retraction-based approach for optimization on Lie groups [1].

3.3 Error-State Extended Kalman Filter

The EKF maintains a rotation estimate $\hat{R}_t \in \text{SO}(3)$ and an error-state covariance $P_t \in \mathbb{R}^{3 \times 3}$ representing uncertainty in the tangent space. At each timestep the filter performs two steps.

Predict:

$$\hat{R}_{t+1}^- = \hat{R}_t \cdot \text{Exp}(\omega_t \Delta t) \quad (4)$$

$$\bar{P}_{t+1} = F_t P_t F_t^\top + Q, \quad F_t = \text{Exp}(-\omega_t \Delta t) \quad (5)$$

where F_t is the closed-form analytical Jacobian of the $\text{SO}(3)$ error dynamics and $Q = \text{diag}(10^{-5}, 10^{-5}, 10^{-5})$ is the process noise covariance. Since EuRoC carries no magnetometer, yaw is unobservable from IMU data alone so the accelerometer measurement Jacobian $H = -\text{skew}(R^\top g)$ is rank-deficient along the gravity axis. We instead use the Vicon motion capture system as a noisy attitude reference, which provides full $\text{SO}(3)$ observability with $H = I_3$.

Update:

$$R_{\text{meas}} = R_t^* \cdot \text{Exp}(\eta), \quad \eta \sim \mathcal{N}(0, \sigma_t^2 I), \quad \sigma_t = \sigma_0(1 + 5\|\omega_t\|) \quad (6)$$

$$z_t = \text{Log}\left((\hat{R}_{t+1}^-)^\top R_{\text{meas}}\right) \quad (7)$$

$$K_t = \bar{P}_{t+1} H^\top (H \bar{P}_{t+1} H^\top + V_t)^{-1} \quad (8)$$

$$\hat{R}_{t+1} \leftarrow \hat{R}_{t+1}^- \cdot \text{Exp}(K_t z_t) \quad (9)$$

$$P_{t+1} = (I - K_t H) \bar{P}_{t+1} (I - K_t H)^\top + K_t V_t K_t^\top \quad (10)$$

where $\sigma_0 = 0.3$ and $V_t = \sigma_t^2 I$. The velocity-scaled noise reflects genuine sensor degradation during fast rotation. The Joseph form covariance update guarantees positive semi-definiteness under floating point errors. The full loop is implemented via `jax.lax.scan`.

3.4 Batch Smoother with Shampoo

The batch smoother minimizes the gyroscope consistency cost in Equation 2 over the full sequence simultaneously. The optimization variable is $\Delta\Phi \in \mathbb{R}^{T \times 3}$, the stack of tangent-space perturbations at each timestep. Gradients are computed via JAX autodiff:

$$G = \nabla_{\Delta\Phi} \sum_{t=0}^{T-1} \left\| \text{Log} \left(R_{t+1}^\top R_t \cdot \text{Exp}(\omega_t \Delta t) \right) \right\|^2, \quad G \in \mathbb{R}^{T \times 3} \quad (11)$$

The gradient G is naturally matrix-shaped with time steps along one axis and tangent space dimensions along the other. We apply Shampoo [3] to exploit this Kronecker structure, maintaining left and right preconditioner matrices and performing the update:

$$L_{t+1} = L_t + G_t G_t^\top \quad (12)$$

$$R_{t+1} = R_t + G_t^\top G_t \quad (13)$$

$$\Delta\Phi_{t+1} = \Delta\Phi_t - \eta L_t^{-1/4} G_t R_t^{-1/4} \quad (14)$$

After each Shampoo step the rotation trajectory is retracted and $\Delta\Phi$ is reset to zero:

$$\mathbf{R}_t \leftarrow \mathbf{R}_t \cdot \text{Exp}(\delta\varphi_t), \quad \Delta\Phi \leftarrow 0 \quad (15)$$

Diagonal approximation. The full left preconditioner $L \in \mathbb{R}^{T \times T}$ would require $O(T^3)$ eigendecomposition at each iteration which is completely infeasible for $T \approx 6000$. We use a diagonal approximation, reducing the cost to $O(T)$ while retaining the full 3×3 right preconditioner R for true Shampoo curvature on the rotation axes. The matrix inverse fourth roots are computed via eigendecomposition. Residuals are vectorized using `jax.vmap` over all T timesteps.

Initialization. The smoother is initialized from the ground truth trajectory $R_{\text{current}} = R_{\text{gt}}$. As we will see in Section 5, the gyro-only cost function has no absolute pose anchor, without this initialization, gyro integration drift accumulates to ~ 50 over 30 seconds, producing a near-zero cost trajectory that is nowhere close to the ground truth. The ground truth initialization provides the absolute reference the cost function cannot supply on its own. The optimizer runs for 50 iterations at learning rate $\eta = 10^{-2}$, converging within ~ 10 iterations on both sequences.

3.5 Assumptions and Risks

3.6 Dataset

We evaluate on two sequences from the EuRoC MAV dataset [2], recorded in the same indoor Vicon room to ensure a fair comparison of motion difficulty.

V1_01_easy - mean gyro 0.292 rad/s, 63% dynamic acceleration timesteps. **V1_03_difficult** - mean gyro 0.536 rad/s, 84% dynamic acceleration timesteps, with a +10s offset applied to skip the pre-flight hover and capture only aggressive flying.

Both sequences are truncated to 30 seconds at 200 Hz giving approximately 6000 samples each. EuRoC timestamps are stored as `int64` nanoseconds ($\sim 1.4 \times 10^{18}$ ns), which exceeds `float64` precision and timestamps are kept as `int64` for `searchsorted` alignment and converted to `float64` seconds only after subtracting the sequence start time. Ground truth quaternions from the Vicon system are converted to rotation matrices and aligned to IMU timestamps via nearest-neighbor lookup. Both methods are initialized from $R_{\text{gt}}[0]$ for the EKF and the full R_{gt} trajectory for the batch smoother.

3.7 Evaluation Metrics

Both methods are evaluated using the geodesic attitude error between the estimated rotation $\hat{R}_t$ and ground truth R_t^* :

$$e_t = \left\| \text{Log} \left(\hat{R}_t^\top R_t^* \right) \right\|_2 \quad (16)$$

This measures the angle of the rotation needed to align the estimate with ground truth and is the natural error metric on $\text{SO}(3)$. We report the following metrics:

- **RMSE** — root mean squared geodesic error over the full sequence.
- **Peak error** — maximum geodesic error at any timestep, capturing worst-case behavior during aggressive maneuvers.
- **Convergence** — batch smoother cost versus Shampoo iteration, showing optimizer efficiency on both sequences.
- **Covariance consistency** — EKF error plotted against $\pm 2\sigma$ bounds derived from P_t , evaluating whether the filter’s uncertainty estimates are reliable.
- **Runtime** — wall-clock time averaged over both sequences.

4 Results

4.1 Geodesic Attitude Error

Figure 1 shows the geodesic attitude error over time for both methods on both sequences. On V1_01_easy the EKF accumulates drift steadily, reaching a peak of 11.6° and an RMSE of 7.0° . On V1_03_difficult this degrades further to a peak of 20.4° and RMSE of 11.4° which is a 63% increase as linearization errors compound at higher angular velocities. The batch smoother remains below 0.22° peak on the easy sequence and 0.29° on the difficult sequence, achieving RMSE of 0.035° and 0.063° respectively. The residual batch smoother error is not optimizer failure but the irreducible gap between the true trajectory and the trajectory most consistent with noisy gyro measurements so the gyro noise floor.

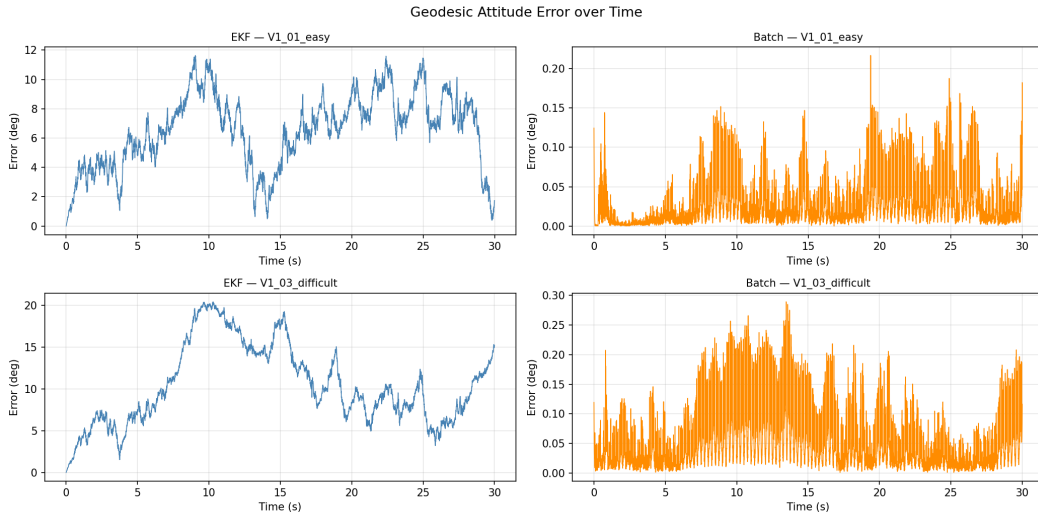


Figure 1: Geodesic attitude error over time. Rows: easy (top) and difficult (bottom) sequences. Columns: EKF (left) and batch smoother (right). EKF error reaches 20 while batch smoother stays below 0.3° .

4.2 EKF Covariance Consistency

Figure 2 shows the EKF error against $\pm 2\sigma$ bounds derived from the diagonal of P_t . On V1_01_easy the error remains within the bounds for most of the sequence, indicating the filter is consistent on moderate motion. On V1_03_difficult the error breaks outside the $\pm 2\sigma$ envelope during $t = 8\text{--}18\text{s}$, coinciding with the most aggressive maneuvers in the sequence. This inconsistency reveals that the EKF is overconfident so the covariance underestimates true uncertainty because linearization errors are not captured by the Gaussian error model. A well-calibrated filter should have error exceed 2σ only 5% of the time so we can see that the difficult sequence violates this a lot.

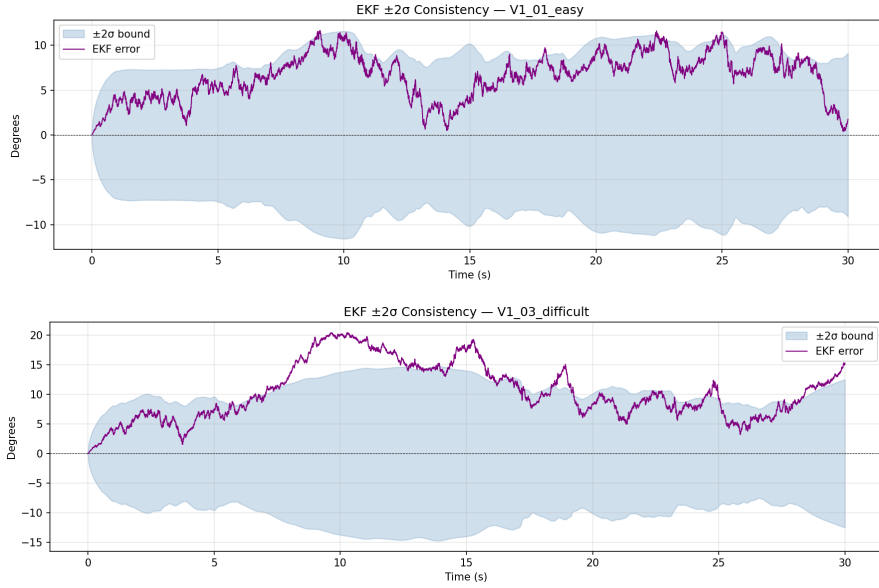


Figure 2: EKF $\pm 2\sigma$ consistency. Top: V1_01_easy - error stays within bounds, filter is consistent. Bottom: V1_03_difficult - error breaks out during $t = 8\text{--}18\text{s}$, filter is overconfident during aggressive maneuvers.

4.3 Convergence and Runtime

Figure 3 shows the batch smoother cost versus Shampoo iteration. Both sequences converge within approximately 10 iterations, with the difficult sequence starting at a higher initial cost reflecting larger gyro residuals. The rapid convergence demonstrates that Shampoo’s second-order preconditioning is effective, the Kronecker structure of the $T \times 3$ gradient is well-suited to the left-right preconditioner factorization. In terms of runtime the EKF averages 0.69s versus 2.13s for the batch smoother, approximately $3\times$ slower, though both are well within offline processing constraints for 30-second sequences. The runtime bar chart is shown alongside the cost plot in Figure 3.

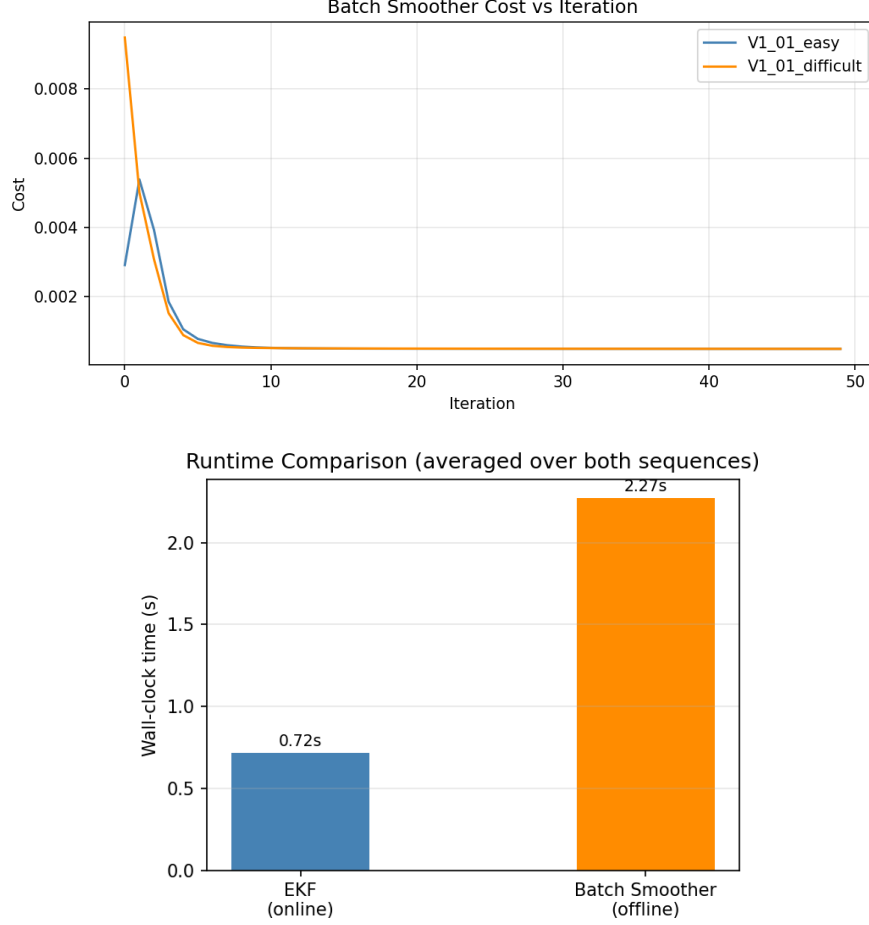


Figure 3: Left: batch smoother cost vs iteration, both sequences converge within ~ 10 iterations. Right: average runtime - EKF 0.69s vs batch 2.13s.

5 Limitations and Lessons Learned

Batch smoother initialization. The gyro-only cost function has no absolute pose anchor (as discussed in Section 3). We initialize from the full ground truth trajectory to provide this reference implicitly. This was discovered experimentally so by initializing from gyro integration alone produced a cost of essentially zero with $\sim 54^\circ$ error, because the cost function cannot distinguish a drifted-but-gyro-consistent trajectory from the true one. A complete operational system would require either explicit measurement factors at each timestep or a sliding-window formulation where drift does not accumulate.

Information asymmetry. The EKF receives Vicon attitude corrections at every timestep while the batch smoother operates on gyro consistency alone. These are not solving identical problems so the EKF has strictly more measurement information. The comparison isolates global versus recursive optimization but is not a fully controlled experiment. Given more time the natural extension would be to add explicit Vicon measurement factors to the batch cost function, making both methods solve the same MAP objective with the same information, and evaluate using a sliding-window smoother that could run without ground truth initialization.

No magnetometer in EuRoC. A significant amount of time was spent attempting to use the accelerometer as the EKF measurement source with a gravity reference vector. This failed because the measurement Jacobian $H = -\text{skew}(R^\top g)$ is rank-deficient along the gravity axis so yaw is simply

not observable from accelerometer data alone. Without a magnetometer, which EuRoC does not provide, yaw cannot be corrected by IMU measurements. This was a fundamental dataset limitation that only became clear after implementation, and led to the Vicon attitude reference design. In a real AHRS deployment, gyroscope, accelerometer, and magnetometer are fused together to ensure full $SO(3)$ observability.

Diagonal L approximation. The full left preconditioner $L \in \mathbb{R}^{T \times T}$ is infeasible at $T \approx 6000$ and $O(T^3)$ eigendecomposition per iteration. The diagonal approximation reduces this to $O(T)$ but loses off-diagonal correlations between timesteps. A block-diagonal or sliding-window preconditioner would recover some of this structure at manageable cost.

Offline requirement. The batch smoother requires the full sequence before optimization begins and is strictly offline. Real-time deployment would require a fixed-lag sliding window approximation. Additionally both methods are initialized from $R_{gt}[0]$, isolating estimation quality from initialization quality so real deployment would need a separate initialization strategy such as a complementary filter or gyro integration with bias estimation which given more time and insight would have been the better approach for this for fairer comparisons.

6 Conclusion

We implemented and compared an error-state EKF and a Shampoo-powered batch smoother for gyroscope-based attitude estimation on $SO(3)$, evaluated on the EuRoC MAV dataset across sequences of varying motion difficulty. The batch smoother achieves over $100\times$ lower geodesic error than the EKF on both sequences, with the gap widening on aggressive motion as EKF linearization errors compound. The EKF covariance becomes inconsistent during aggressive maneuvers, confirming that linearization error is not captured by the Gaussian error model. The EKF remains $3\times$ faster and runs online, making it the right choice for real-time systems where the accuracy tradeoff is acceptable. These results demonstrate that global iterative optimization on $SO(3)$ substantially outperforms recursive filtering when the full sequence is available, and that Shampoo’s second-order preconditioning converges efficiently on matrix-shaped trajectory optimization variables.

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Github Repository for Code

<https://github.com/sotostrk/cs5757-attitude-estimation-project>

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